cp_7a

1

Print N Times | 25/25

Easy

Write a program that reads a number `N` and prints the number `0` `N` times on `N` lines.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` lines containing an integer `0` on each line.

Explanation

For example, if the given number is `N = 4` ,

- The number `0` should be printed `4` times.

The output should be,

```
0
0
0
0
```

Sample Input 1

```
4
```

Sample Output 1

```
0
0
0
0
```

Sample Input 2

```
2
```

Sample Output 2

```
0
0
```

code:

```
n=int(input())
count=0
while count<n:
```

```
    print("0")
    count=count+1__
```

2

Print N Integers | 30/30

Write a program to print integers from 1 to the given integer (N).

Input

The first line of input will contain a positive integer.

Output

The output should be of N lines, printing an integer in each line.

Explanation

For example, if the given number is 3, your code should print the

```
1
2
3
```

Similarly, if the given number is 5, your code should print the

```
1
2
3
4
5
```

Sample Input 1

```
3
```

Sample Output 1

```
1
2
3
```

Sample Input 2

```
5
```

Sample Output 2

```
1
2
3
4
5
```

code:

```
n=int(input())
nums=1
while nums<=n:
    print(nums)
    nums=nums+1
```

3

First N Natural Numbers | 35/35

Easy

Write a program that reads a number `N` and prints `N` Natural Numbers from **1**.

Note

All the positive numbers from **1** are called **Natural Numbers**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should contain `N` Natural Numbers from **1**, each on a new line.

Explanation

For example, if the given number is `N = 3` ,

- The **3** numbers from **1** are **1**, **2** and **3**.

The output should be,

```
1
2
3
```

Sample Input 1

```
3
```

Sample Output 1

```
1
2
3
```

Sample Input 2

```
5
```

Sample Output 2

```
1
2
3
4
5
```

code:

```
num=int(input())
n=1
while n<=num:
    print(n)
    n=n+1
```

4

Print Integers - 2 | 30/30

Given two integer numbers M and N, write a program to print the integers from M to N.

Input

The first line of input will contain an integer.
The second line of input will contain an integer.

Output

The output should be printing an integer in each line, starting from M to N.

Explanation

For example, if the given numbers are 2 and 6, the output should be

```
2
3
4
5
6
```

Similarly, if the given numbers are -2 and 2, the output should be

```
-2
-1
0
1
2
```

Sample Input 1

```
2
6
```

Sample Output 1

```
2
3
4
5
6
```

Sample Input 2

```
-2
2
```

Sample Output 2

```
-2
-1
0
1
2
```

code:

```
n1=int(input())
n2=int(input())
diff=n2-n1
```

```
count=0
while count<=diff:
    print(n1)
    n1=n1+1
    count=count+1
```

```
5
```

Average of the N Numbers | 30/30

Easy

Write a program that reads a number `N` and prints the average of `N` numbers from **1**.

Note

The average of `N` numbers from **1** can be calculated as,

Average = Sum of `N` numbers from **1** / Count of numbers (`N`)

Example: If `N = 3` , the average of **3** numbers from **1**
Average = (1 + 2 + 3) / 3 = 2.0

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a float that is the average of `N` numbers from **1**.

Explanation

For example, if the given number is `N = 4` ,

- The **4** numbers from **1** are **1**, **2**, **3** and **4**.
- The sum of the numbers is **10**. (`1 + 2 + 3 + 4 = 10`)
- The average of the numbers is **2.5**. (`10 / 4 = 2.5`)

The output should be **2.5**.

Sample Input 1

```
4
```

Sample Output 1

```
2.5
```

Sample Input 2

```
7
```

Sample Output 2

```
4.0
```

code:

```
num=int(input())
avg=(1+num)/2
print(avg)
```

6

Cubes of N Numbers | 30/30

Easy

Write a program that reads a number `N` and prints the cube of `N` numbers from **1**.

Note

The cube of a number `X` is `X` to the power of **3**. (X^3)

Input

The input will be a single line containing an integer representing `N` .

Output

The output should contain cubes of `N` numbers from **1**, each on a new line.

Explanation

For example, if the given number is `N = 4` ,

- The **4** numbers from **1** are **1**, **2**, **3** and **4**.
- The cube of the numbers is **1**, **8**, **27** and **64**.

The output should be,

```
1
8
27
64
```

Sample Input 1

```
4
```

Sample Output 1

```
1
8
27
64
```

Sample Input 2

```
5
```

Sample Output 2

```
1
8
27
64
```

code:

```
n=int(input())
count=1
while count<=n:
    print(count**3)
    count=count+1
```

7

Solid Rectangle | 35/35

Solved

Given two integers M and N, write a program to print a solid rectangle pattern of M rows and N columns using the asterisk character (*).

Input

The first line of input will contain a positive integer.
The second line of input will contain a positive integer.

Output

The output should be the M number of lines and N number of columns with the asterisk character (*).
Note: There is a space after each asterisk character(*).

Explanation

For example, if the given numbers of rows are 2 and the number of columns is 3, the output should be 2 lines and 3 columns with the asterisk character(*).

```
* * *
* * *
```

For example, if the given numbers of rows are 4 and the number of columns is 6, the output should be 4 lines and 6 columns with the asterisk character(*).

```
* * * * * *
* * * * * *
* * * * * *
* * * * * *
```

Sample Input 1

```
2
3
```

Sample Output 1

```
* * *
* * *
```

Sample Input 2

```
4
6
```

Sample Output 2

```
* * * * * *
* * * * * *
* * * * * *
* * * * * *
```

code:

```
n=int(input())
n2=int(input())
count=1
while count<=n:
    print("* "*n2)
    count=count+1
```

8

10 Numbers after N | 30/30

Easy

Write a program that reads a number `N` and prints 10 numbers after `N` .

Input

The input will be a single line containing an integer representing `N` .

Output

The output should contain 10 numbers after `N` , each on a new line.

Explanation

For example, if the given number is `N = 2` ,

- The 10 numbers after 2 are 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12.

The output should be,

```
3
4
5
6
7
8
9
10
11
12
```

Sample Input 1

```
2
```

Sample Output 1

```
3
4
5
6
7
8
9
10
11
12
```

code:

```
n=int(input())
n_10=n+10
count=n
while count<n_10:
```

```
    n=n+1
    print(n)
    count=count+1
```

9

N Numbers after M | 50/50

Easy

Write a program that reads two numbers `M` and `N` and prints `N` numbers after `M` .

Input

The first line of input contains an integer representing `M` .
The second line of input contains an integer representing `N` .

Output

The output should contain `N` numbers after `M` , each on a new line.

Explanation

For example, if the given numbers are `M = 3` and `N = 5` ,

- The **5** numbers after **3** are **4**, **5**, **6**, **7** and **8**.

The output should be,

```
4
5
6
7
8
```

Sample Input 1

```
3
5
```

Sample Output 1

```
4
5
6
7
8
```

Sample Input 2

```
11
2
```

Sample Output 2

```
12
13
```

code:

```
m=int(input())
n=int(input())
n_m=n+m
count=m
while count<n_m:
    m=m+1
    print(m)
    count=count+1
```

10

Sum of N Numbers from M | 30/30

Easy

Write a program that reads two numbers `M` and `N` and prints the **sum** of `N` numbers from `M` .

Input

The first line of input contains an integer representing `M` .
The second line of input contains an integer representing `N` .

Output

The output should be a single line containing an integer that is the **sum** of `N` numbers from `M` .

Explanation

For example, if the given numbers are `M = 7` and `N = 3` ,

- The **3** numbers from **7** are **7**, **8** and **9**.
- The **sum** of **3** numbers from **7** is **24**. (`7 + 8 + 9 = 24`)

Number	Previous Sum	Previous Sum + Number = Current Sum
7	0	0 + 7 = 7
8	7	7 + 8 = 15
9	15	15 + 9 = 24

The output should be **24**.

Sample Input 1

```
7
3
```

Sample Output 1

```
24
```

Sample Input 2

```
2
10
```

Sample Output 2

```
65
```

code:

```
n=int(input())
m=int(input())
sum=0
count=1
while count<=m:
    sum=sum+n
```

```
    n=n+1
    count=count+1
print(sum)
```

11

Solid Right Angled Triangle | 40/40

Given an integer number (N) as input. Write a program to print the right-angled triangular pattern of N lines using an asterisk(*) character.

Input

The first line of input will contain a positive integer.

Output

The output should be N lines with an asterisk(*) character printing in a right-angled triangular pattern.
Note: There is a space after each asterisk(*) character.

Explanation

For example, if the given number is 4,
the output should be

```
*
* *
* * *
* * * *
```

For example, if the given number is 2,
the output should be

```
*
* *
```

Sample Input 1

```
4
```

Sample Output 1

```
*
* *
* * *
* * * *
```

Sample Input 2

```
2
```

Sample Output 2

```
*
* *
```

code:

```
n=int(input())
count=1
star_count=1
while count<=n:
    star="* "
    print(star*star_count)
    star_count=star_count+1
    count=count+1
```

12

Sum of N Natural Numbers | 40/40

Write a program that reads a number `N` and prints the **sum** of `N` Natural Numbers.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the **sum** of `N` Natural Numbers.

Explanation

For example, if the given number is `N = 6` ,

- The numbers from **1** to **6** are **1**, **2**, **3**, **4**, **5** and **6**.
- The **sum** of the numbers is **21**. (`1 + 2 + 3 + 4 + 5 + 6 = 21`)

Number	Previous Sum	Previous Sum + Number = Current Sum
1	0	0 + 1 = 1
2	1	1 + 2 = 3
3	3	3 + 3 = 6
4	6	6 + 4 = 10
5	10	10 + 5 = 15
6	15	15 + 6 = 21

The output should be **21**.

Sample Input 1

```
6
```

Sample Output 1

```
21
```

Sample Input 2

```
3
```

Sample Output 2

```
6
```

code:

```
n=int(input())
sum=0
count=1
while count<=n:
    sum=sum+count
```

```
    count=count+1
print(sum)
```

13

Read N Inputs | 40/40 ● Solved

Given an integer N, write a program which reads N inputs and prints them.

Input

The first line of input will contain a positive integer, N.
The following N lines will contain an integer in each line.

Output

The output should be N lines, containing an integer per line.

Explanation

For example, if the given N is 3, then read the inputs in the next 3 lines and print them. If the given input integers in the next three lines are 8, 11, and 25, the output should be

```
8
11
25
```

Sample Input 1

```
3
8
11
25
```

Sample Output 1

```
8
11
25
```

Sample Input 2

```
2
7
20
```

Sample Output 2

```
7
20
```

code:

```
n=int(input())
count=1
while count<=n:
    st=int(input())
```

```
    print(st)
    count=count+1
```

14

Product of N Inputs | 35/35

Medium

Given a number `N` , write a program that reads `N` numbers as input and prints the **product** of the given `N` numbers.

Input

The first line of input contains an integer representing `N` .
The next `N` lines of input contain integers.

Output

The output should be the **product** of the given `N` inputs.

Explanation

For example, if the given number is `N = 3` ,

- The **3** inputs are **2**, **3**, and **7**.
- The **product** of **2**, **3**, and **7** is **42**. (`2 * 3 * 7 = 42`)

Number	Previous Product	Previous Product * Number = Current Product
2	1	1 * 2 = 2
3	2	2 * 3 = 6
7	6	6 * 7 = 42

The output should be **42**.

For example, if the given number is `N = 4` ,

- The **4** inputs are **11**, **2**, **4**, and **9**.
- The **product** of **11**, **2**, **4**, and **9** is **792**. (`11 * 2 * 4 * 9 = 792`)

Number	Previous Product	Previous Product * Number = Current Product
11	1	1 * 11 = 11
2	11	11 * 2 = 22
4	22	22 * 4 = 88
9	88	88 * 9 = 792

The output should be **792**.

code:

```
n=int(input())
count=0
prod=1
while count<n:
    num=int(input())
    prod=num*prod
    count=count+1
    if num==0:
        print(0)
print(prod)
```

15

Sum of the Given Numbers | 45/45 — Solved

Given an integer N, write a program which reads N inputs and prints the sum of the given input integers.

Input

The first line of input will contain a positive integer, N.
The next N lines will contain the integers, each in a line.

Output

The output should be the sum of the given input integers.

Explanation

For example, if the given N is 3, then read the inputs in the next 3 lines and print the sum of the three input integers. If the given input integers in the next three lines are 8, 11, and 25, the output should be 44.

Sample Input 1

```
3
8
11
25
```

Sample Output 1

```
44
```

Sample Input 2

```
2
7
20
```

Sample Output 2

```
27
```

code:

```
n=int(input())
sum=0
count=0
while count<n:
    num=int(input())
    num_1=num+1
    sum=sum+num
    count=count+1
print(sum)
```

16

Identify the Mistake - Print Characters | 10/10

Write a program that reads a word and prints each character of the word in a new line.

Input

The first line of input will contain a word.

Output

The output should contain N lines, equal to the length of the word. Each of the N lines should contain a character of the word.

Explanation

For example, if the given word is "Python".
Your code should print each character of the word in a line.
So the output should be

```
P
y
t
h
o
n
```

Sample Input 1

```
Python
```

Sample Output 1

```
P
y
t
h
o
n
```

Sample Input 2

```
Loops
```

Sample Output 2

```
L
o
o
p
s
```

code:

```python
a = input()
counter = 0
length_of_a = len(a)
while counter <= (length_of_a - 1):
    print(a[counter])
    counter = counter + 1
```